Alec Stanley

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Melbourne, VIC | Australian Citizen

EDUCATION

Bachelor of Engineering (Honours) and Bachelor of Science, Monash University – Clayton, VIC Expected Dec 2027

- Majors in Aerospace Engineering and Astrophysics; completed minors in Physics and Mathematics
- Distinction average (WAM 77.1); High Distinctions in CFD, spacecraft dynamics, astrophysics and programming
- Certified SolidWorks Associate (CSWA), scored 235/240, May 2025

Ursula Frayne Catholic College – Perth, WA | ATAR 98.65 2022

SKILLS

Programming: Python (NumPy, Matplotlib, multiprocessing), MATLAB, C, C++, Mathematica, Arduino, Git

Simulation and Analysis: Computational fluid dynamics (SU2), large-eddy simulation, aerodynamics, compressible flow, boundary layer theory, rigid-body and attitude dynamics, numerical integration, thermodynamics, photometry

Design and Fabrication: SolidWorks (CSWA certified), 3D printing, laser cutting, electronics and custom PCB assembly

PROJECTS

N-Body Gravitational Simulator – C++, High Distinction June 2026

- Built a gravitational simulator in C++ from scratch: a physics integration engine, plotting engine and window manager
- Implemented linked lists, vectors and arrays as custom data stores with manual memory management

Boundary Layers in Laminar and Turbulent Flow Regimes – CFD study, High Distinction Apr 2026

- Simulated laminar and turbulent flat-plate boundary layers at Re 1.3×10^6 and 10×10^6, plus compressible flow over a 2D cylinder, in the open-source SU2 solver
- Validated thicknesses and skin friction against Blasius theory, locating flow separation at 126.8°

Galaxy Collision Simulation – Python, High Distinction Oct 2025

- Integrated 242 gravitating bodies with a Leapfrog scheme to model the observed M51 and NGC 5195 collision, conserving total energy with angular momentum error at order 10^{-12}
- Cut runtime from tens of minutes to seconds using pre-allocated arrays, rendering 3 animations in parallel

Glider Design, Build and Flight Performance – Design Build Fly team project Oct 2025

- Developed a low-Reynolds glider over 3 iterations in a team of 5, using an Eppler E423 airfoil at aspect ratio 8
- Modelled lift, drag and stability in MATLAB, predicting a maximum glide ratio of 14.95, then compared to flight tests

Tumbling Rocket, Intermediate Axis Theorem – MATLAB, High Distinction, report 10/10 Oct 2025

- Propagated torque-free rotation of an arbitrary STL geometry, reproducing the Dzhanibekov intermediate-axis instability
- Contrasted stable roll and yaw against chaotic 180° pitch reversals to justify spacecraft spin-axis choice

Autonomous Rover – 38th Warman National Design and Build Competition May 2025

- Developed the electronics, control code and CAD in a team of 5 for a rover collecting 3 objects in 120 seconds
- Redesigned the drivetrain around mecanum wheels, a string-and-pulley telescoping arm and 2 ultrasonic sensors
- Built a custom Arduino control board and fabricated the chassis from laser-cut timber and 3D-printed parts

EXPERIENCE

Resident Advisor, Richardson Hall, Monash University – Clayton, VIC Feb 2024 – June 2025

- Responded to after-hours welfare and behavioural incidents across a community of 200 residents
- Coordinated community events and mediated conflict, escalating sensitive cases under duty-of-care procedures

Service Supervisor, Coles Supermarkets – Perth, WA and Melbourne, VIC Nov 2019 – Present

- Supervised front-end teams across 2 stores through peak trading, de-escalating customer and staff conflict
- Managed inventory, daily financial reporting and point-of-sale systems while training new team members